SemGuard: Semantics-Decoupled Watermarking for Gaussian Splatting

Yingying Shi, Zhong Zhou, Bin Zhou

***Abstract*—3D Gaussian Splatting has emerged as a powerful representation for high-fidelity and real-time 3D/4D content creation, raising the need for effective copyright protection of Gaussian assets. Recent CLIP-guided watermarking methods offer an efficient solution by learning universal message decoders from textual watermark representations and applying them to rendered visual features. However, we identify an important yet overlooked limitation of this paradigm: the decoder is trained on watermark-centric textual features but deployed on semantics-entangled visual features, where weak watermark signals are mixed with geometry, appearance, illumination, background, viewpoint, and motion. This representation mismatch, which we term the semantic entanglement gap, makes scene semantics an implicit nuisance factor during watermark extraction and degrades robustness across diverse scenarios. We propose SemGuard, a semantics-decoupled watermarking framework for Gaussian Splatting. Instead of training the decoder only on clean watermark features or adapting it for each Gaussian asset, SemGuard explicitly injects visual semantics as controllable interference during decoder pre-training. Specifically, we fuse watermark features encoded by the frozen CLIP text encoder with visual semantic features extracted by the frozen CLIP image encoder, thereby constructing semantics-entangled representations that mimic the decoder inputs encountered during watermark extraction. The decoder is then required to recover the original message from these mixed features, encouraging it to identify watermark-induced patterns under semantic interference. We further introduce progressive semantic interference curriculum, which gradually increases interference strength to stabilize training and improve semantic-invariant message extraction. Experiments on diverse 3D and 4D Gaussian Splatting benchmarks demonstrate that SemGuard achieves superior decoding accuracy, semantic robustness, and cross-dataset generalization over existing CLIP-guided watermarking methods while preserving visual fidelity and embedding efficiency. Code is available at https://github.com/shisyy/SemGuard.**

***Index Terms*—Asset protection, gaussian splatting, watermark decoding, CLIP, semantic disentanglement.**

I. Introduction

3D Gaussian Splatting (3DGS) has recently emerged as a powerful representation for high-fidelity novel-view synthesis and real-time rendering [1]. By representing scenes with explicit anisotropic Gaussian primitives and differentiable splatting, 3DGS enables efficient optimization and compact asset storage. Its extensions to dynamic and 4D Gaussian Splatting further support time-varying 3D content [2]–[4]. As Gaussian assets become increasingly used in graphics, virtual reality, and digital content creation, protecting their copyright becomes an important problem. Digital watermarking provides an effective solution by embedding imperceptible binary messages into digital assets for later ownership verification [5]–[7]. For Gaussian assets, an ideal watermarking method should preserve rendering fidelity, remain robust under viewpoint changes and post-processing distortions, and generalize across diverse scenes and motions. Recent works have explored watermarking neural radiance fields and Gaussian Splatting assets [8]–[12]. Among them, CLIP-guided methods provide an efficient paradigm by learning a universal message decoder in the CLIP embedding space [12], [13]. Specifically, binary messages are encoded as CLIP text features and decoded by a lightweight decoder, which is then frozen and reused to extract messages from CLIP visual features of watermarked Gaussian renderings. This decoupled design avoids scene-specific decoder optimization and improves scalability.

However, this decoupled paradigm introduces a fundamental limitation in universal decoder learning. As illustrated in Fig. 1, existing CLIP-guided methods train the decoder using watermark-centric text features but deploy it on visual features extracted from rendered Gaussian views. During pre-training, the decoder mainly observes message-related patterns encoded by the CLIP text encoder. After deployment, however, the decoder receives image features containing both weak watermark signals and diverse visual semantics, including object appearance, background, viewpoint, and temporal variations. Although CLIP provides a shared vision-language space, it primarily aligns high-level semantic concepts, whereas watermark extraction requires recovering non-semantic binary information. This issue is particularly challenging for Gaussian watermarking because the embedded signals are intentionally designed to be imperceptible, making them much weaker than dominant semantic variations captured by CLIP. Consequently, the decoder suffers from a training–inference representation mismatch, where semantic content interferes with watermark evidence. We refer to this mismatch as the semantic entanglement gap, which causes unstable message extraction across different Gaussian assets, viewpoints, and motions. A straightforward solution is to adapt the decoder for each Gaussian asset during watermark optimization. However, this converts the universal decoder into a scene-specific extractor, increasing optimization cost and sacrificing the scalability of CLIP-guided watermarking. Therefore, the semantic entanglement gap should be addressed during decoder pre-training before deployment.

In this paper, we propose SemGuard, a semantics-decoupled

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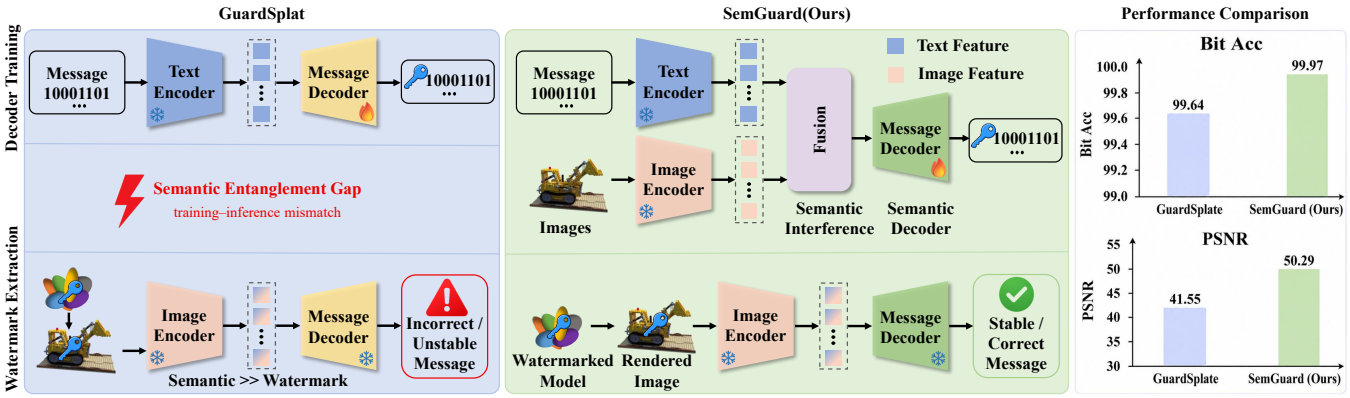


Fig. 1. Motivation and overview of SemGuard. Existing CLIP-guided Gaussian watermarking learns a universal decoder from watermark-centric textual features, while watermark extraction relies on visual features entangled with scene semantics, including appearance, viewpoint, and motion, leading to a training–inference semantic entanglement gap. SemGuard resolves this gap by injecting visual semantics as controllable interference during decoder learning, producing a semantic-robust decoder that enables reliable watermark extraction from diverse 3D and 4D Gaussian assets.

watermarking framework for Gaussian Splatting. Instead of adapting the decoder to individual assets, SemGuard learns a semantic-robust universal decoder by explicitly introducing visual semantic interference during decoder pre-training. Specifically, given a binary message, we encode its CLIP text representation as the watermark feature and fuse it with visual semantic features extracted by the frozen CLIP image encoder. The resulting mixed representation simulates the semantics-entangled features encountered during watermark extraction. The decoder is then trained to recover the original message from these mixed features, encouraging it to focus on watermark-related information under semantic variations. To further stabilize decoder learning, we introduce progressive semantic interference curriculum, which gradually increases semantic interference during training. The decoder first learns message recovery under weak interference and progressively adapts to stronger visual variations. After pre-training, the decoder is frozen and directly integrated into the standard Gaussian watermark embedding pipeline, where only compact SH offsets are optimized. Therefore, SemGuard improves decoder robustness without sacrificing the efficiency and reusability of CLIP-guided watermarking.

We evaluate SemGuard on diverse 3D and 4D Gaussian Splatting benchmarks. Experiments demonstrate that SemGuard achieves improved watermark extraction accuracy, robustness, and visual fidelity compared with existing Gaussian watermarking methods. These results verify that explicitly modeling semantic interference during decoder learning is critical for building reusable and robust watermark extractors. Our contributions are summarized as follows:

- We reveal the semantic entanglement gap in CLIP-guided Gaussian watermarking, where a text-trained universal decoder is deployed on semantics-entangled visual features.
- We propose SemGuard, a semantic-robust watermarking framework that learns a semantic-robust universal decoder by injecting visual semantic interference during pre-training. We further introduce progressive semantic interference curriculum for stable decoder learning.
- Extensive experiments on 3D and 4D Gaussian Splatting benchmarks demonstrate that SemGuard improves watermark extraction accuracy and robustness while preserving rendering fidelity.

II. Related Work

A. Neural and Gaussian Scene Representations

Neural scene representations have significantly advanced novel-view synthesis and digital asset creation. Neural Radiance Fields (NeRF) represent scenes as continuous volumetric radiance fields and achieve high-quality view synthesis from posed images [14]. However, NeRF-based methods rely on computationally expensive volumetric sampling [15]–[17], limiting their efficiency for interactive applications. 3D Gaussian Splatting (3DGS) addresses this limitation by representing scenes with explicit anisotropic Gaussian primitives equipped with opacity and spherical-harmonic appearance coefficients [1]. Through differentiable splatting and visibility-aware rasterization, 3DGS achieves real-time rendering while maintaining high visual fidelity. Recent works further extend Gaussian representations to improve geometry quality, anti-aliasing, and dynamic scene modeling [18], [19]. Dynamic Gaussian representations model time-varying content through temporal deformation and appearance changes [2]–[4], [20], enabling high-quality 4D asset creation. However, dynamic Gaussian assets also introduce new protection challenges, as embedded watermark signals must remain stable under viewpoint changes, temporal deformation, and motion-induced appearance variations.

B. Digital Watermarking for 2D and Neural Assets

Digital watermarking aims to embed imperceptible information into digital content for copyright protection, ownership verification, and provenance tracing. Classical watermarking methods typically modify transform-domain representations, such as DCT, DWT, or SVD coefficients, to improve robustness against compression and image processing operations [5], [21]–[23]. With the development of deep learning, neural watermarking methods jointly optimize message embedding

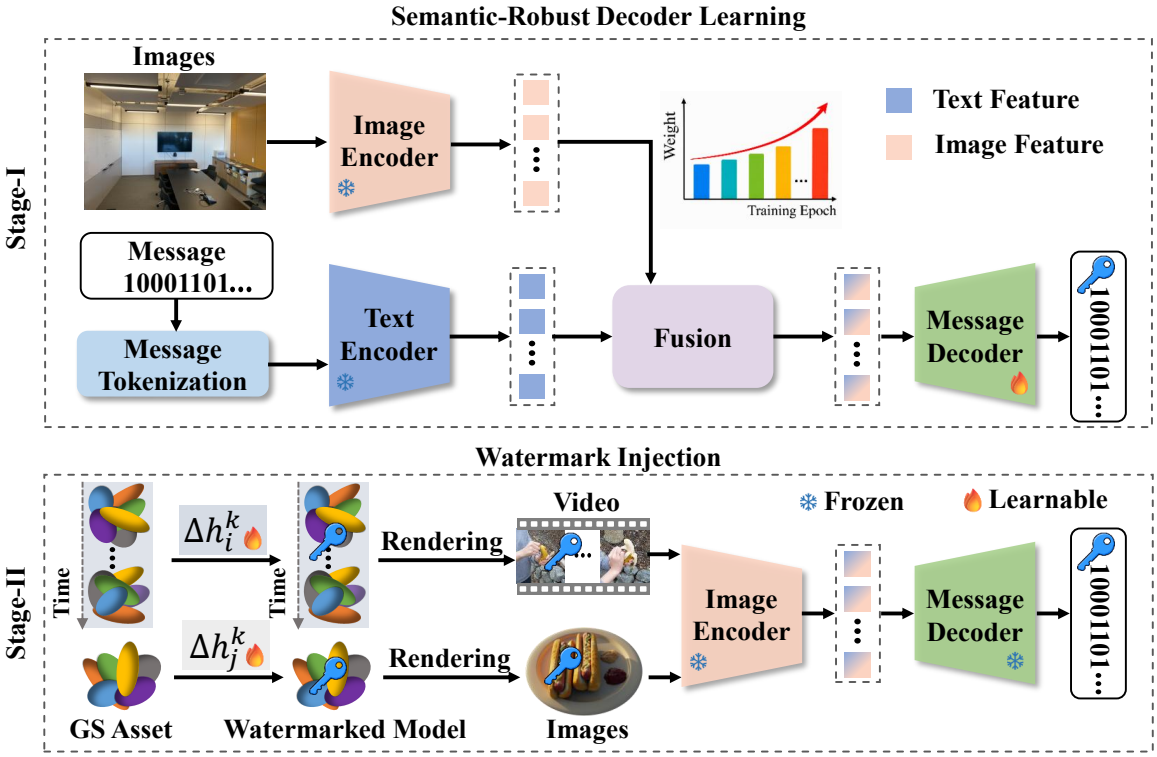


Fig. 2. Overview of SemGuard. SemGuard consists of two stages: semantic-robust decoder learning and SH-offset-based Gaussian watermark embedding. In Stage I, CLIP textual watermark features are combined with visual semantic features to train a robust decoder under semantics-entangled representations. In Stage II, the frozen decoder guides SH-offset optimization for embedding messages into 3D/4D Gaussian assets, and the embedded message is recovered from rendered images or videos.

and extraction. HiDDeN introduces an end-to-end encoder-decoder framework for hiding binary messages in images [6], while StegaStamp improves robustness under physical-world distortions [7]. Subsequent works further improve robustness through distortion-aware training, self-supervised latent representations, and invertible mechanisms [24]–[26]. Watermarking has also been extended to generative models and diffusion-based image synthesis [27], [28]. Beyond 2D images, recent works investigate watermarking for 3D assets and neural scene representations. Early approaches mainly focus on meshes and point clouds [29]–[35].

C. Watermarking Gaussian Splatting and CLIP-guided Decoding

Recent works have explored copyright protection for Gaussian Splatting assets. GaussianMarker introduces uncertainty-aware watermarking by perturbing Gaussian parameters while considering rendering uncertainty [10]. GS-Hider embeds messages by replacing SH features with secured representations and jointly optimizing the scene and message decoders [36]. 3D-GSW proposes a robust 3DGS watermarking framework using frequency-guided densification, gradient masking, and wavelet-domain optimization [11]. GuardSplat introduces an efficient CLIP-guided watermarking paradigm that first learns a universal message decoder from CLIP textual features and then embeds messages into SH offsets of Gaussian primitives [12]. By decoupling decoder learning from asset optimization, GuardSplat avoids scene-specific decoder training and improves watermarking scalability.

III. Method

Our goal is to embed a binary copyright message into Gaussian Splatting assets while preserving rendering fidelity and enabling reliable extraction from rendered views under diverse semantic and temporal variations. SemGuard addresses this goal through a semantics-decoupled framework with two key components. First, we train a semantic-robust universal decoder in the CLIP embedding space [13]. Unlike existing CLIP-guided Gaussian watermarking methods that train the decoder solely on watermark-centric textual features [12], SemGuard explicitly introduces visual semantics as controllable interference during decoder learning. Second, the learned decoder is frozen and reused for watermark embedding by optimizing compact SH offsets while keeping Gaussian geometry and opacity fixed. As shown in Fig. 2, SemGuard consists of two stages: semantic-robust decoder learning and SH-offset Gaussian watermark embedding.

A. Gaussian Splatting Preliminaries

A Gaussian Splatting asset represents a scene with a set of anisotropic Gaussian primitives [1]. We denote the asset as

$$\mathcal{G} = \{\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, \alpha_i, \mathbf{h}_i\}_{i=1}^{N}, \tag{1}$$

where $\boldsymbol{\mu}_i$ is the center of the i-th Gaussian, $\boldsymbol{\Sigma}_i$ is its covariance, α_i is the opacity, and $\mathbf{h}_i$ denotes spherical-harmonic

(SH) coefficients for view-dependent appearance. Following common 3DGS implementations, the SH coefficients are divided into a DC component and higher-order components:

$$\mathbf{h}_i = \{\mathbf{h}_i^{\mathrm{dc}}, \mathbf{h}_i^{\mathrm{rest}}\}. \tag{2}$$

Given a camera pose π, the differentiable renderer produces a view:

$$\mathbf{I} = \mathcal{R}(\mathcal{G}, \pi). \tag{3}$$

For dynamic or 4D Gaussian assets, we use t to denote the temporal index and absorb time-dependent deformation or appearance changes into the renderer [3]:

$$\mathbf{I}_t = \mathcal{R}(\mathcal{G}, \pi, t). \tag{4}$$

Since rendering is differentiable with respect to SH appearance parameters, we optimize compact SH perturbations to embed messages while preserving the original geometry.

B. Semantic-Robust Decoder Learning

Semantic interference-aware decoder learning. We train the decoder under controllable semantic interference by mixing CLIP textual watermark features with CLIP visual features during decoder learning. For each binary message $\mathbf{m}$, we first convert it into a CLIP token sequence $\mathbf{t_m}$ and obtain its textual watermark feature using the frozen CLIP text encoder:

$$\mathbf{F}_T = E_T(\mathbf{t_m}). \tag{5}$$

Meanwhile, we sample an auxiliary image $\mathbf{I}_s$ from a semantic image pool and extract its visual semantic feature using the frozen CLIP image encoder:

$$\mathbf{F}_I = E_V(\mathbf{I}_s). \tag{6}$$

The auxiliary image is independent of the target message and contains no watermark. It only provides semantic variations unrelated to the target message. The concrete image sources are described in the experimental setup. We then construct the decoder input by linearly weighting the textual watermark feature and the visual semantic feature:

$$\mathbf{F}_{mix} = (1-\rho)\mathbf{F}_T + \rho\mathbf{F}_I, \tag{7}$$

where $\rho \in (0, 1)$ is the semantic interference weight. A larger ρ introduces stronger visual semantic variation, making the decoder recover messages from increasingly entangled representations. When $\rho = 0$, the training reduces to conventional text-only decoder learning. As ρ increases, the decoder is encouraged to recover the message from features increasingly influenced by image semantics. The mixed feature is normalized and decoded as:

$$\mathbf{o} = D_\psi\left(\frac{\mathbf{F}_{mix}}{\|\mathbf{F}_{mix}\|_2}\right), \tag{8}$$

where $\mathbf{o} \in \mathbb{R}^L$ denotes the predicted bit logits. The decoder is trained with the standard binary cross-entropy loss with logits:

$$\mathcal{L}_{dec} = \mathrm{BCEWithLogits}(\mathbf{o}, \mathbf{m}). \tag{9}$$

During this stage, both CLIP encoders are frozen and only the message decoder D_ψ is optimized. After training, D_ψ is kept fixed and reused for all Gaussian assets.

Progressive semantic interference curriculum. To stabilize decoder training, we use a continuous weighting schedule for semantic interference. Let e denote the current epoch and E denote the total number of epochs. The semantic interference weight is linearly increased from $\rho_{\min}$ to $\rho_{\max}$:

$$\rho_e = \rho_{\min} + (\rho_{\max} - \rho_{\min}) \cdot \frac{e}{E}. \tag{10}$$

At epoch e, the decoder input is constructed as:

$$\mathbf{F}_{mix}^e = (1-\rho_e)\mathbf{F}_T + \rho_e\mathbf{F}_I. \tag{11}$$

This curriculum allows the decoder to first learn watermark recovery under weak interference and progressively adapt to stronger semantics-entangled representations.

C. SH-offset Gaussian Watermark Embedding

After learning the semantic-robust decoder, we freeze D_ψ and embed a target message by optimizing compact SH offsets in the Gaussian representation [12]. Let $\mathcal{G}$ be the original asset and $\widetilde{\mathcal{G}}$ be the watermarked copy. We keep the Gaussian geometry and opacity fixed, and only modify the SH coefficients. For each Gaussian i and each SH block $k \in \{\mathrm{dc}, \mathrm{rest}\}$, we define:

$$\widetilde{\mathbf{h}}_i^k = \mathbf{h}_i^k + \Delta\mathbf{h}_i^k. \tag{12}$$

All offsets are stacked into $\Delta\mathbf{H}^{\mathrm{dc}}$ and $\Delta\mathbf{H}^{\mathrm{rest}}$. This parameterization preserves the original reconstructed geometry and binds the watermark to the Gaussian representation rather than to a specific rendered image. Since SH coefficients control view-dependent appearance, the watermark signal is distributed across rendered views through differentiable rasterization. Given a sampled camera pose π, we render the watermarked view and its clean reference:

$$\widetilde{\mathbf{I}} = \mathcal{R}(\widetilde{\mathcal{G}}, \pi), \qquad \mathbf{I} = \mathcal{R}(\mathcal{G}, \pi). \tag{13}$$

The frozen CLIP visual encoder and the frozen SemGuard decoder extract message logits from the watermarked rendering:

$$\mathbf{o} = D_\psi\left(E_V(\widetilde{\mathbf{I}})\right). \tag{14}$$

For 4D Gaussian assets, we additionally sample temporal indices:

$$\mathbf{o}_t = D_\psi\left(E_V\left(\mathcal{R}(\widetilde{\mathcal{G}}, \pi, t)\right)\right). \tag{15}$$

The same target message is enforced across all sampled views and frames. During embedding, we optionally apply differentiable post-processing distortions to rendered views, following common Gaussian watermarking pipelines [12]. This augmentation improves robustness to distortions such as cropping, scaling, rotation, brightness changes, and JPEG compression. The distortion layer is used only during optimization.

D. Optimization and Verification

During Gaussian watermark embedding, only the SH offsets are optimized. The CLIP visual encoder E_V and the semantic-robust decoder D_ψ are frozen. Given a batch of B rendered views or frames, the message loss is:

$$\mathcal{L}_{\mathrm{msg}} = \frac{1}{BL}\sum_{b=1}^{B}\sum_{i=1}^{L}\mathrm{BCEWithLogits}(o_{b,i}, m_i). \tag{16}$$

TABLE I

QUANTITATIVE COMPARISON ON 3DGS WATERMARKING BENCHMARKS. WE EVALUATE WATERMARK EMBEDDING AND EXTRACTION ON BLENDER AND LLFF 3DGS ASSETS UNDER DIFFERENT MESSAGE LENGTHS, REPORTING BIT ACC AND RENDERING FIDELITY METRICS (PSNR, SSIM, AND LPIPS). FOR SEMGUARD VARIANTS, SUPERSCRIPTS DENOTE DIFFERENT SEMANTIC INTERFERENCE SOURCES USED DURING STAGE-I DECODER TRAINING: ⋆ USES STATIC SCENE IMAGES FROM BLENDER AND LLFF; † USES BOTH STATIC AND DYNAMIC SCENE IMAGES.

Methods	16 bits				32 bits				48 bits			
	Bit Acc	PSNR	SSIM	LPIPS	Bit Acc	PSNR	SSIM	LPIPS	Bit Acc	PSNR	SSIM	LPIPS
NeRF-based Watermarking Methods												
CopyRNeRF [8]	91.16	26.29	0.9100	0.0380	78.08	26.13	0.8960	0.0410	60.06	27.56	0.8950	0.0660
WateRF [9]	95.67	32.79	0.9480	0.0330	88.58	31.19	0.9360	0.0400	85.82	30.86	0.9300	0.0400
3DGS Built on Watermarked Images												
CIN [26] + 3DGS [1]	56.73	31.95	0.9194	0.1027	53.13	31.74	0.9279	0.0944	55.78	30.25	0.9139	0.1120
SSL [25] + 3DGS [1]	58.94	36.51	0.9737	0.0094	61.85	35.24	0.9706	0.0179	58.79	35.88	0.9710	0.0123
HiDDeN [6] + 3DGS [1]	63.07	31.59	0.9790	0.0171	52.46	34.51	0.9682	0.0209	53.08	33.42	0.9687	0.0299
DwtDctSvd [23] + 3DGS [1]	55.44	34.78	0.9582	0.0399	53.15	32.32	0.9477	0.0547	51.83	31.09	0.9302	0.0704
StegaStamp [7] + 3DGS [1]	79.72	35.52	0.9697	0.0181	82.36	34.04	0.9601	0.0265	83.97	32.54	0.9523	0.0406
3DGS optimized by altering attributes												
3DGS [1] w/ WateRF [9]	92.89	31.01	0.9678	0.0475	90.15	29.56	0.9611	0.0492	87.30	29.13	0.9562	0.0534
3D-GSW [11]	-	-	-	-	97.37	35.08	0.978	0.043	93.72	33.31	0.970	0.045
GaussianMarker [10]	99.36	34.42	0.9820	0.0124	98.85	33.98	0.9788	0.0163	98.25	32.12	0.9723	0.0234
GuardSplat [12]	99.64	41.55	0.9957	0.0017	99.04	39.40	0.9939	0.0022	98.29	38.90	0.9923	0.0028
SemGuard with different semantic interference sources												
SemGuard(Ours⋆)	**99.97**	**50.29**	**0.9993**	**0.0004**	**99.47**	**43.59**	**0.9975**	**0.0011**	**98.72**	**42.32**	**0.9969**	**0.0014**
SemGuard(Ours†)	**99.91**	**46.97**	**0.9986**	**0.0007**	**99.52**	**43.72**	**0.9975**	**0.0011**	**98.05**	**42.58**	**0.9969**	**0.0014**

To preserve visual fidelity, we combine pixel-level, structural, and perceptual reconstruction losses between the watermarked rendering $\widetilde{\mathbf{I}}$ and the clean reference $\mathbf{I}$ [37], [38]:

$$\mathcal{L}_{\mathrm{rgb}} = (1-\lambda_{\mathrm{ssim}})\|\widetilde{\mathbf{I}}-\mathbf{I}\|_1 + \lambda_{\mathrm{ssim}}\left(1-\mathrm{SSIM}(\widetilde{\mathbf{I}},\mathbf{I})\right), \quad (17)$$

$$\mathcal{L}_{\mathrm{recon}} = \mathcal{L}_{\mathrm{rgb}} + \lambda_{\mathrm{lpips}}\mathrm{LPIPS}(\widetilde{\mathbf{I}},\mathbf{I}). \quad (18)$$

We further regularize the SH offsets:

$$\mathcal{L}_{\mathrm{off}} = \|\Delta\mathbf{H}^{\mathrm{dc}}\|_2^2 + \|\Delta\mathbf{H}^{\mathrm{rest}}\|_2^2. \quad (19)$$

The final objective is:

$$\mathcal{L} = \lambda_{\mathrm{msg}}\mathcal{L}_{\mathrm{msg}} + \lambda_{\mathrm{recon}}\mathcal{L}_{\mathrm{recon}} + \lambda_{\mathrm{off}}\mathcal{L}_{\mathrm{off}}. \quad (20)$$

This objective balances accurate message extraction, rendering fidelity, and compact watermark perturbation.

For verification, the owner keeps the target message $\mathbf{m}$ and the private decoder D_ψ. Given a suspicious Gaussian asset, the verifier renders K views or frames $\{\mathbf{I}_k^{sus}\}_{k=1}^K$ and averages the decoded logits:

$$\bar{\mathbf{o}} = \frac{1}{K}\sum_{k=1}^{K} D_\psi\left(E_V(\mathbf{I}_k^{sus})\right). \quad (21)$$

The final message is obtained by thresholding $\sigma(\bar{\mathbf{o}})$, and ownership is verified by the bit accuracy between the extracted message and the target message.

IV. EXPERIMENTS

A. Dataset

We use rendered images from static and dynamic Gaussian scene datasets as semantic interference sources for Stage-I decoder learning, and evaluate watermarking performance on corresponding 3D/4D Gaussian assets in Stage-II. Specifically, Blender [14] and LLFF [40] provide static scene semantics, while D-NeRF [41] and HyperNeRF [42] provide dynamic appearance and motion variations. These images are only used to extract CLIP visual features for decoder training and contain no watermark information. For Gaussian watermark embedding and evaluation, we follow existing Gaussian watermarking protocols and evaluate static watermarking on Blender and LLFF 3DGS assets, as well as dynamic watermarking on D-NeRF and HyperNeRF 4DGS assets.

B. Evaluation Metrics

We evaluate SemGuard from three aspects: watermark extraction, visual fidelity, and robustness. For extraction, we report Bit Accuracy (Bit Acc) under different message lengths and semantic training sources. For rendering quality, we use PSNR, SSIM [37], and LPIPS [38]. For robustness, we evaluate Bit Acc under common distortions, including noise, rotation, scaling, blur, cropping, brightness changes, JPEG compression, VAE attacks, and combined distortions. For 4DGS assets, metrics are averaged over sampled viewpoints and temporal frames.

C. Implementation Details

We use CLIP ViT-B/32 [13] as the frozen visual and textual encoder. For Stage-I decoder learning, only the lightweight message decoder is optimized. For Stage-II watermark embedding, the decoder and CLIP image encoder are frozen, and only SH offsets are optimized while keeping Gaussian geometry, opacity, and deformation parameters fixed. We train decoders for different message lengths separately and optimize SH offsets with Adam. The detailed training configurations and hyperparameters are provided in the supplementary material.

TABLE II

QUANTITATIVE COMPARISON ON 4DGS WATERMARKING BENCHMARKS. WE EVALUATE WATERMARK EMBEDDING AND EXTRACTION ON D-NERF AND HYPERNERF 4DGS ASSETS UNDER DIFFERENT MESSAGE LENGTHS, REPORTING BIT ACC AND RENDERING FIDELITY METRICS (PSNR, SSIM, AND LPIPS). FOR SEMGUARD VARIANTS, SUPERSCRIPTS DENOTE DIFFERENT SEMANTIC INTERFERENCE SOURCES USED DURING STAGE-I DECODER TRAINING: △ USES DYNAMIC SCENE IMAGES FROM D-NERF AND HYPERNERF; † USES BOTH DYNAMIC AND STATIC SCENE IMAGES.

Methods	16 bits				32 bits				48 bits			
	Bit Acc	PSNR	SSIM	LPIPS	Bit Acc	PSNR	SSIM	LPIPS	Bit Acc	PSNR	SSIM	LPIPS
Gaussian-based watermarking methods												
GuardSplat [12]	87.61	43.82	0.9972	0.0014	76.06	40.35	0.9947	0.0029	72.61	38.98	0.9935	0.0039
SemGuard with different semantic interference sources												
SemGuard(Ours△)	**92.75**	**47.23**	**0.9985**	**0.0008**	**90.67**	**44.05**	**0.9975**	**0.0014**	**87.89**	**43.35**	**0.9972**	**0.0015**
SemGuard(Ours†)	**95.35**	**47.39**	**0.9987**	**0.0010**	**91.91**	**45.41**	**0.9981**	**0.0011**	**86.85**	**43.31**	**0.9971**	**0.0015**

TABLE III

ABLATION STUDY ON THE SEMANTIC INTERFERENCE CURRICULUM. WE COMPARE FIXED INTERFERENCE WEIGHTS $\rho \in \{0.1, 0.3, 0.5, 0.7\}$ WITH PROGRESSIVE INTERFERENCE SCHEDULES ON 3DGS BENCHMARKS. BOLD VALUES INDICATE THE BEST PERFORMANCE IN EACH COLUMN.

Strategy	16 bits				32 bits				48 bits			
	Bit Acc	PSNR	SSIM	LPIPS	Bit Acc	PSNR	SSIM	LPIPS	Bit Acc	PSNR	SSIM	LPIPS
Fixed interference weights												
0.1	99.95	47.10	0.9987	0.0004	98.17	39.91	0.9942	0.0028	95.81	38.25	0.9923	0.0036
0.3	99.93	46.27	0.9984	0.0008	98.02	40.58	0.9951	0.0023	98.12	40.25	0.9950	0.0024
0.5	99.96	48.23	0.9990	0.0005	98.80	41.14	0.9958	0.0020	98.59	40.66	0.9955	0.0021
0.7	99.96	**50.56**	0.9993	0.0004	99.21	42.59	0.9967	0.0015	98.70	41.88	0.9963	0.0017
Progressive interference schedules												
Step	99.96	48.10	0.9989	0.0005	99.18	42.59	0.9969	0.0014	98.37	42.23	0.9968	0.0014
Continuous	**99.97**	50.29	**0.9993**	**0.0004**	**99.42**	**43.61**	**0.9975**	**0.0011**	**98.72**	**42.32**	**0.9969**	**0.0014**

TABLE IV

ROBUSTNESS COMPARISON UNDER POST-PROCESSING DISTORTIONS. WE EVALUATE 16-BIT WATERMARK EXTRACTION ON 3DGS BENCHMARKS AND REPORT BIT ACCURACY (%). THE SUPERSCRIPTS ⋆ AND † DENOTE SEMANTIC INTERFERENCE SOURCES USED DURING STAGE-I DECODER TRAINING: ⋆ USES STATIC SCENE IMAGES, WHILE † ADDITIONALLY USES DYNAMIC SCENE IMAGES.

Methods	**Bit Acc. (%) ↑**									
	None	**Noise** (μ=0.1)	**Rotation** ($\pm\pi/6$)	**Scaling** ($\leq 25\%$)	**Blur** (σ=0.1)	**Crop** (40%)	**Brightness** (0.5∼1.5)	**JPEG** (10% **quality**)	**VAE Attack [39]**	**Combined**
CopyRNeRF [8]	91.6	90.04	88.13	89.33	90.06	–	–	–	–	–
WateRF [9]	95.67	95.36	93.13	93.29	95.25	95.40	90.91	86.99	51.73	84.12
3DGS [1] w/ WateRF [9]	92.89	87.35	88.28	90.33	91.92	89.07	88.71	88.49	55.48	86.37
GaussianMarker [10]	99.36	99.13	70.84	97.89	94.40	98.52	95.78	86.22	52.00	83.49
3D-GSW [11]	97.37	83.84	87.94	94.64	96.01	92.86	–	92.65	–	90.84
GuardSplat [12]	99.64	99.60	94.56	98.75	99.27	98.71	97.46	94.70	82.35	93.38
SemGuard(Ours⋆)	99.97	99.81	99.01	99.98	99.94	99.65	99.44	99.53	96.26	98.24
SemGuard(Ours†)	99.91	99.79	98.86	99.56	99.90	99.54	99.39	99.48	96.04	97.65

D. Comparison with Other Methods

Results on static 3DGS benchmarks. Table I compares SemGuard with NeRF-based watermarking, image-based watermarking combined with 3DGS, and recent Gaussian watermarking methods on Blender and LLFF. Across different message lengths, SemGuard achieves the best trade-off between extraction accuracy and rendering fidelity. Compared with NeRF-based methods and image watermarking approaches, SemGuard obtains substantially higher Bit Acc, especially for longer messages. This is because SemGuard directly embeds watermark signals into the Gaussian representation through SH offsets, avoiding the inconsistency of preserving image-level watermarks during 3D reconstruction. Compared with Gaussian watermarking methods, SemGuard further improves rendering fidelity while maintaining near-perfect extraction accuracy. For example, under 16-bit messages, SemGuard(⋆) improves GuardSplat from 99.64% to 99.97% Bit Acc, while increasing PSNR from 41.55 dB to 50.29 dB and reducing LPIPS from 0.0017 to 0.0004. These improvements demonstrate that the semantic-robust decoder provides more stable optimization signals for SH-offset watermark embedding. Different semantic interference sources lead to comparable performance. SemGuard(⋆), trained with Blender and LLFF images, achieves the best results on static benchmarks due to its closer semantic distribution. SemGuard(†), which further includes dynamic images, maintains competitive performance, showing that broader semantic variations do not harm decoder generalization.

Results on dynamic 4DGS benchmarks. Table II evaluates SemGuard on D-NeRF and HyperNeRF. Compared with GuardSplat, SemGuard consistently improves watermark extraction and rendering fidelity under different message lengths.

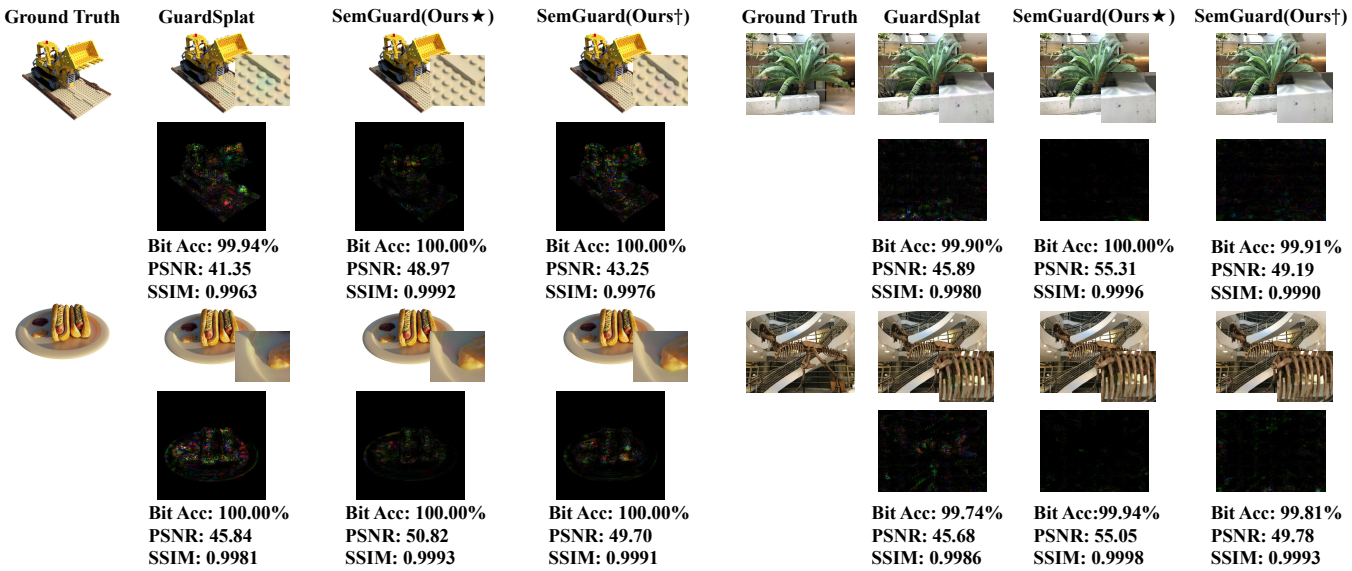


Fig. 3. Overview of SemGuard. Visual comparison on representative Blender 3DGS scenes and LLFF 3DGS scenes with 16-bit messages. We compare GuardSplat [12] with SemGuard variants trained with different semantic interference sources. The error maps visualize rendering differences, where darker regions indicate smaller perturbations.

For example, SemGuard(△) improves 16-bit Bit Acc from 87.61% to 92.75% and PSNR from 43.82 dB to 47.23 dB. For 48-bit messages, it still achieves 87.89% Bit Acc with 43.35 dB PSNR, demonstrating robustness under temporal deformation and dynamic appearance changes. The results indicate that semantic-robust decoder learning transfers well from static semantic interference to dynamic Gaussian assets. Adding static semantic sources to dynamic semantic interference further improves extraction performance for shorter messages while maintaining competitive performance for longer messages. Overall, these experiments validate that modeling semantic interference before decoder deployment improves watermark extraction across both static and dynamic Gaussian representations.

E. Ablation Study

Ablation study on the semantic interference curriculum. Table III studies different interference weighting strategies. Increasing fixed interference weights generally improves extraction and fidelity, indicating that stronger semantic variation helps reduce the mismatch between decoder training and watermark extraction. Compared with fixed weights, progressive strategies further improve performance. The continuous schedule achieves the best overall results, reaching 99.97%, 99.42%, and 98.72% Bit Acc for 16-, 32-, and 48-bit messages, respectively. This confirms that gradually increasing semantic interference provides an effective curriculum, allowing the decoder to first learn message recovery and then adapt to semantics-entangled representations.

F. Robustness against Distortions

Table IV evaluates watermark robustness under common post-processing distortions. SemGuard consistently outperforms previous methods across almost all attacks. For example, under rotation, JPEG compression, and VAE attacks, SemGuard(⋆) improves Bit Acc from 94.56%, 94.70%, and 82.35% of GuardSplat to 99.01%, 99.53%, and 96.26%, respectively. Different semantic interference sources also maintain strong robustness. The static semantic source achieves the best performance on most individual distortions, while broader semantic sources provide comparable or improved generalization under combined attacks. These results demonstrate that semantic interference-aware decoder learning improves robustness beyond clean-view extraction.

G. Qualitative comparison.

Figure 3 compares rendered views and error maps between GuardSplat and SemGuard variants on Blender and LLFF 3DGS assets. SemGuard preserves visual appearance closer to the original Gaussian assets while maintaining accurate message extraction. The darker and sparser error maps indicate that SemGuard introduces smaller appearance perturbations during SH-offset optimization. This improvement is attributed to the semantic-robust decoder learned in Stage I, which provides more reliable watermark extraction signals and enables more effective SH-offset optimization.

SemGuard achieves consistently high Bit Accuracy across different object-centric and scene-level assets, demonstrating that the learned decoder generalizes beyond specific visual contents. In particular, most Blender and LLFF scenes maintain over 99% Bit Acc for 16-bit messages, showing that the decoder is not sensitive to specific object appearances or scene layouts. For 16-bit messages, most scenes achieve nearly perfect extraction accuracy, with Bit Acc close to 100%. As the message length increases, extraction becomes more challenging because more bits need to be recovered from the limited watermark capacity. Nevertheless, SemGuard maintains stable performance across different scenes, while preserving high rendering fidelity with low LPIPS values. Comparing different semantic interference sources, SemGuard(⋆) trained with static

scene images generally achieves stronger performance on static 3DGS assets. This indicates that semantic interference closer to the deployment distribution provides more effective decoder supervision. SemGuard(†), which additionally introduces dynamic semantic sources, still maintains competitive performance, showing that broader semantic variations do not degrade decoder generalization.

V. Conclusion

We presented SemGuard, a semantics-decoupled watermarking framework for Gaussian Splatting that improves the reliability of universal watermark extraction from rendered Gaussian assets. Through our analysis of CLIP-guided watermarking, we identified a fundamental representation mismatch between decoder learning and watermark extraction: the decoder is optimized with watermark-centric textual representations but applied to visual representations containing diverse asset semantics. This semantic entanglement gap limits the robustness of existing universal decoders when facing different scenes, viewpoints, and temporal variations. SemGuard addresses this limitation by shifting the focus from scene-specific decoder adaptation to semantic-robust decoder learning. By exposing the decoder to controllable visual variations before deployment, SemGuard enables the decoder to distinguish watermark evidence from semantic content while preserving the scalability of CLIP-guided watermarking. The learned decoder can be directly integrated into Gaussian watermark embedding, where compact SH offsets provide an efficient way to embed messages without modifying the underlying geometry. Extensive experiments on static and dynamic Gaussian Splatting benchmarks demonstrate that SemGuard consistently improves watermark extraction accuracy and robustness while maintaining high rendering fidelity. These results highlight the importance of considering the deployment distribution during universal decoder learning and suggest that explicitly modeling nuisance variations is a promising direction for robust watermarking of emerging neural assets.

In future work, we will investigate broader semantic and temporal variations for decoder pre-training and explore more general watermarking strategies for diverse neural representations beyond Gaussian Splatting.

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